Full-State Estimation

Our system

$$\begin{cases} \boxed{\dot{x} = Ax + Bu} \\ \boxed{y = Cx} \end{cases}$$

$x \in \mathbb{R}^n$
$u \in \mathbb{R}^q$
$y \in \mathbb{R}^p$

controllable on A, B.
observable on A, C.
Then
we can estimate $\hat{x}(t)$ from measurement $y(t)$

$$\text{i)}\quad \frac{d}{dt}\hat{x} = A\hat{x} + Bu + \underbrace{K_f}_{\text{Kalman Filter Gain Matrix}}(y - \hat{y})$$

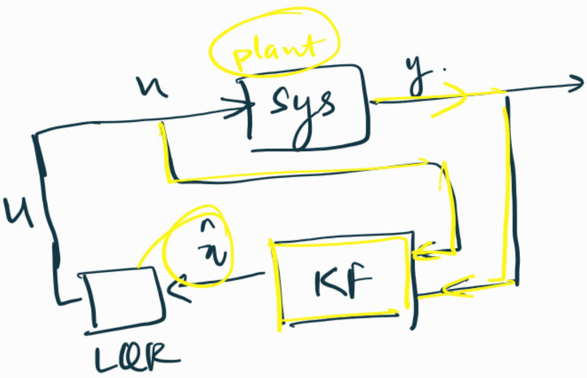
$$\text{ii)}\quad \hat{y} = C\hat{x}$$

$$\Rightarrow \frac{d}{dt}\hat{x} = A\hat{x} + Bu + K_f y - K_f C\hat{x}$$

$$= (A - K_f C)\hat{x} + [B \;\; K_f]\begin{bmatrix} u \\ y \end{bmatrix}$$

$$\dot{\hat{x}} = A'\hat{x} + B'u'$$

↳ New dynamical system



Computing Error.

$$\varepsilon = x - \hat{x} \Rightarrow \frac{d\varepsilon}{dt} = \frac{dx}{dt} - \frac{d\hat{x}}{dt}$$

$$= (Ax + \cancel{Bu}) - \left\{ (A - K_f C)\hat{x} + [B \;\; K_f]\begin{bmatrix} \cancel{u} \\ y \end{bmatrix} \right\}$$

$$= A\underbrace{(x - \hat{x})}_{\varepsilon} + K_f(C\hat{x} - y)$$

$$K_f C(\hat{x} - x)$$

$$= (A - K_f C)\cancel{(\hat{x} - x)}\,\varepsilon.$$

choose. $\hat{\varepsilon}$ / poles.
↳ $z_1 \;\; z_2$
$K_f = K_f(z_1, z_2)$

$$\Rightarrow \dot{\varepsilon} = (A - K_f C)\varepsilon.$$

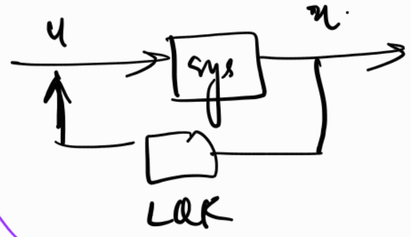
Algebraic-Riccati Equation.
↳ infinite Horizon problem

LQR $\dot{x} = Ax + Bu$.

if $(A, B) \rightarrow$ controllable

$\downarrow$

Solve for K



$u = -Kx$.

$\dot{x} = Ax - BKx$

$\boxed{\dot{x} = (A - BK)x} \rightarrow$ Riccati Equation.

$n \rightarrow \text{Rank}(A)$

Controllability

$\left[A \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B \right]$

Rank $\rightarrow$ full rank